WEEK 3 · DAY 3 · RULE BOOK

The Slow Creep: Drift and CUSUM

Follow it step by step. Every common error and its fix is here.

Your goal today: write a CUSUM detector, catch the amplifier's step and a secret fault, find when the battery started ageing faster, and earn **14 green ✅ ticks**. Time: about 60 minutes. Keep this page open next to Colab.

Before you start (2 minutes)

☐ Open `W3D3-exercise` in Colab, then **File → Save a copy in Drive**. Work only in the copy.

☐ Run **STEP 0**. Wait for ✅ `Ready!`. It also loads three helpers: `expected_temp`, `fade_rate`, `rolling_slope`.

☐ Run every cell **top to bottom**. Part B needs your CUSUM from Part A.

Today's 6 golden rules

#	Rule	Why
1	**Name the shape first**: spike, level shift or drift	Each shape needs a different tool
2	**Take away what you already know** (season, drive level, ageing) before you watch	Otherwise the normal yearly swing hides the fault
3	**CUSUM line:** `S = max(0, S + leftover − k)`	Add today, minus the slack, never below zero
4	**Set k and h from σ**: `k = 0.5 * sigma`, `h = 5 * sigma`	σ = the normal wobble of the leftovers in a calm period
5	**Alarm date ≠ start date.** Start = the day after S was last 0	CUSUM needs time to collect evidence
6	**Drift? Watch the slope, not the level**	The battery never crosses 95 Ah, but its fade gets 1.6× faster

The exercise, step by step

Task	What you do	The tool you need
A1	Name three shapes	`"spike"`, `"level shift"`, `"drift"`
A2	Write the CUSUM line	`max(0, S + x − k)`

Task	What you do	The tool you need
A3	Start day and alarm day of the toy example (h = 2)	read the printed table
A4	What is k for?	`"a"`, `"b"` or `"c"` in quotes
B1	Days above the whole-series 3σ ceiling	`(tw["twta_temp_c"] > ceiling).sum()`
B2	The step in units of local wobble	`(post.mean() - pre.mean()) / pre.std()`
B3	Leftovers after 12 May 2025	`expected_temp(tw)`, then `resid["2025-05-12":].mean()`
B4	The two knobs	`0.5 * sigma` and `5 * sigma`
B5	First CUSUM alarm in 2025	`S[S > h].index.min()`
B6	Secret fault: alarm date and start date	`Sp[Sp > h].index.min()` and `last_zero + pd.Timedelta(days=1)`
C1	Days below 95 Ah	`(cap < 95).sum()`
C2	Speed before and after April 2025	`fade_rate(cap[:"2025-03-31"])`, `fade_rate(cap["2025-04-01":])`
C3	First day the rolling slope crosses the warning line	`WARNING_LINE = -3.0`
🌶️ **Bonus**	False alarms with a jumpy h = 2σ	count alarm days before `start`

The CUSUM recipe

```
def cusum(values, k):
    S = 0
    out = []
    for x in values:              # x = today's leftover (actual - expected)
        S = max(0, S + x - k)     # add it, take away the slack, never below zero
        out.append(S)
    return out

S = pd.Series(cusum(resid.to_numpy(), k), index=resid.index)
first_alarm = S[S > h].index.min()                    # first day S is above h
last_zero = S[:first_alarm][S[:first_alarm] == 0].index.max()
start = last_zero + pd.Timedelta(days=1)              # best guess of when the change
began
```

When things go wrong: error → fix

You see	It means	Fix
❌ on A2 with `None` in the list	the blank in the loop is still `___`	write `S = max(0, S + x - k)`
`TypeError: '>' not supported … NoneType`	a blank above is still empty	fill A2 / B4 first, then re-run
`AssertionError: ✍️ Do … first`	a cell needs an earlier answer	do the task it names, then run again
`NameError: name 'sigma' is not defined`	B3 didn't run successfully	fill `expected = expected_temp(tw)` and run B3
CUSUM alarms on day 1 of January	you forgot `- k`, or used `h` where `k` belongs	k = 0.5σ goes **inside** the loop; h is only the alarm line
B6 start date is one day early	you gave `last_zero`, not the day after	`last_zero + pd.Timedelta(days=1)`
Battery "fires" or looks different	wrong file version	use the default `data_path("battery_health_daily")` (week-04)
Everything is weird	–	**Runtime → Restart session and run all**

🏁 Final project · Checkpoint 2: 📥 Data in hand

Bring these to your trainer today (on **your** data, on NIGCOMSAT systems, not Colab):

- ☐ Your unit head's **email reply** agreeing the process, the data and who it's for
- ☐ Your data loaded with a `DatetimeIndex`, one time zone, and `.asfreq()`
- ☐ Profiled: **rows, date range, gaps, duplicated timestamps**
- ☐ One honest plot with your **operational threshold** drawn on it
- ☐ No raw data on GitHub; nothing restricted pasted into an AI tool

Next checkpoint: **Week 4 · Day 3: method drafted.**

Stuck? Climb the ladder

1. **Read the ❌ hint.**
2. **Check the error table above.**
3. **Ask your neighbour.**
4. **Ask the AI button** to *explain the error*: paste code and error, never real data.
5. **Raise your hand.**

🏁 **Finished?** Run `scoreboard()`. 14 of 14 = 🏆. Then try the 🌶️ bonus.